Decent Coaching Classes

X SB

Mind Your Math VIII

Subject : Mathematics Part II **Max. Marks : 40**

Note :

(1) All questions are compulsory.

(2) Use of calculator is not allowed.

(3) The numbers to the right of the questions indicate full marks.

(4) In case of MCQs, write the alphabet of the correct answer only. For activities, rewrite the completed activity.

Q. 1 (A) Four alternative answers are given for every sub-question. Select the correct alternative. (4)

(i) The ratio of the areas of two similar triangles is 16 : 25. The ratio of their corresponding sides is

(A) 16 : 25 (B) 4 : 5 (C) 5 : 4 (D) 8 : 5

(ii) The legs of a right-angled triangle are 9 cm and 12 cm. Its hypotenuse is

(A) 21 cm (B) 18 cm (C) 15 cm (D) 13 cm

(iii) The radius of a circle is 5 cm. The length of the tangent from a point 13 cm from the centre is

(A) 12 cm (B) 8 cm (C) 18 cm (D) 14 cm

(iv) The distance of the point (3, 4) from the origin is

(A) 7 (B) 1 (C) 12 (D) 5

Q. 1 (B) Solve the following sub-questions. (4)

(i) If $\sin\theta = 3/5$, find $\cos\theta$.

(ii) Find the slant height of a cone whose radius is 3 cm and height is 4 cm.

(iii) In a 30°–60°–90° triangle, the hypotenuse is 10 cm. Find the length of the side opposite to the 30° angle.

(iv) The measure of an inscribed angle in a circle is 40°. Find the measure of its intercepted arc.

Q. 2 (A) Complete any two activities and rewrite them. (4)

(i) In ΔABC, points D and E lie on sides AB and AC respectively and DE ∥ BC. If AD = 4, DB = 6 and AE = 6, find EC.

In ΔABC, DE ∥ BC ∴ AD/DB = _____ (Basic proportionality theorem)

∴ 4/6 = 6/_____ ∴ EC = _____

(ii) In ΔPQR, ∠Q = 90°, PQ = 6 and QR = 8. Find PR.

$PR^2 = PQ^2$ + _____ (Pythagoras theorem)

∴ $PR^2 = 36$ + _____ = _____ ∴ PR = _____

(iii) Find the coordinates of the midpoint M of the segment joining A(2, 3) and B(8, 9).

$x = (x_1 + x_2)/2 = (2 +$ _____$)/2 =$ _____

$y = (y_1 + y_2)/2 = (3 + 9)/2 =$ _____

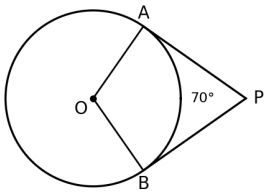
∴ M(_____ , _____)

Q. 2 (B) Solve any two sub-questions. (4)

(i) Find the value of $\sin 60° \times \cos 30° + \tan 45°$.

(ii) The radius of a cylinder is 7 cm and its height is 10 cm. Find its curved surface area. ($\pi = 22/7$)

(iii) In the figure, PA and PB are tangents to the circle with centre O from the external point P. If ∠APB = 70°, find ∠AOB.



Q. 3 (A) Complete any one activity and rewrite it. (3)

(i) ΔABC ~ ΔPQR, A(ΔABC) = 36 sq. unit, A(ΔPQR) = 64 sq. unit and AB = 9 units. Find PQ.

$A(\Delta ABC)/A(\Delta PQR) = AB^2/$_____ (Theorem on areas of similar triangles)

$\therefore\ 36/64 =$ _____$/PQ^2$ $\therefore\ PQ^2 =$ (_____ × 64)/36 = _____ $\therefore$ PQ = _____ units

(ii) Chords AB and CD of a circle intersect at point P inside the circle. If AP = 4, PB = 6 and CP = 3, find PD and CD.

AP × PB = CP × _____ (Theorem of internal division of chords)

$\therefore$ 4 × _____ = 3 × PD $\therefore$ PD = _____

CD = CP + PD = 3 + _____ = _____

Q. 3 (B) Solve any two sub-questions. (6)

(i) The diagonals of a rhombus are 16 cm and 12 cm. Find the side of the rhombus and its perimeter.

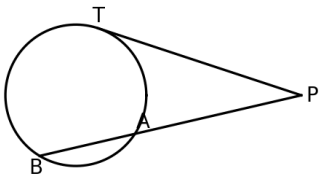
(ii) Find the coordinates of the point P which divides the segment joining A(1, −2) and B(4, 7) internally in the ratio 1 : 2.

(iii) Prove that: $(1 + \tan^2\theta) \times \cos^2\theta = 1$.

Q. 4 Solve any three sub-questions. (9)

(i) In ΔABC, ray AD bisects ∠BAC and D lies on side BC. If AB = 10 cm, AC = 14 cm and BC = 6 cm, find BD and DC.

(ii) In the figure, PT is a tangent to the circle at T and PAB is a secant. If PT = 12 cm and PA = 8 cm, find AB.

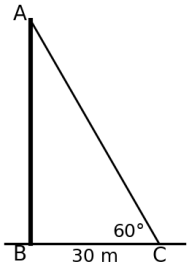


(iii) Using slopes, show that the points A(−2, −3), B(1, 0) and C(4, 3) are collinear.

(iv) The radius of a circle is 6 cm. Find the length of the arc and the area of the sector corresponding to a central angle of 60°. (π = 3.14)

Q. 5 Solve any two sub-questions. (6)

(i) In the figure, AB is a vertical tower. From a point C on the ground, 30 m away from the foot B of the tower, the angle of elevation of the top A is 60°. Find the height of the tower. (√3 = 1.73)



(ii) The radius of a cone is 6 cm and its height is 8 cm. Find (a) its slant height, (b) its curved surface area, (c) its total surface area. (π = 3.14)

(iii) In ΔPQR, ∠Q = 90° and QS ⊥ PR. If PS = 4 cm and SR = 9 cm, find QS and PQ.

— End of Question Paper —